Iran Blocks Signal
Iran has blocked access to Signal after its boost in popularity this month; can be accessed using VPNs. **http://dgit.in/feb21-51**

AllTap
Pine Labs has launched AllTap app for contactless merchant payments and financial services. **http://dgit.in/feb21-52**

funny, but it is just that people used to bikes expect to have a gearbox! "A gearbox is part and parcel of motorcycling and is part of its emotional appeal. By preserving it, we offer a familiar ride experience, thereby lowering mental barriers. Shifting to EVs is a big enough change for most people. By keeping it familiar, we make it easier for people to shift from petrol to electric."

Emote Electric had to create the gearbox from scratch, "There are quite a few additional challenges of creating an electric geared bike. For one, we had to design and develop the gearbox from scratch, since a gearbox from a petrol motorcycle could not do justice to an electric drivetrain. This added a lot of expense, time and complexity to the project as we went from one moving part to nearly 30 and they all had to play well together. Being so new and untried, we also did not have prior design data to rely on and had to fall back on core engineering principles and logic to navigate the developmental process."

For fuelling up, Emote Electric is opting for the option of swappable batteries, which is along the lines of Niti Aayog's vision for electric transportation in India. The idea is that the batteries are charged in stacks at fuel pumps, and swapped out with the vehicles when they come in for juice, with the battery from the bike plugged in for more charging. The approach reduces wait times for recharging electric vehicles. "Swapping stations/kiosks are the general requirement where users can quickly add in more batteries or swap a depleted battery for a fresh one. There can be some flexibility about how the swapping station is managed. The key challenge is the Capex required and the being able to effectively manage assests (i.e the battery) such that a battery is always available for any person who arrives requiring a swap, while still being able to minimise the size of inventory required to meet this requirement."

The bike is also of course, smart, and comes with a companion app that let's you do all kinds of tech wizardry, "the companion app can Lock/unlock the bike remotely, Geo fence it, push navigation directions to the bike, view live location, view telemetry, schedule maintenance, automatically remind the user of maintenance to be carried out, keep a backup of maintenance records and other important data related to the motorcycle."

DIGANTARA TACKLES SPACE DEBRIS

The experience gained by working on student satellites has pushed the young entrepreneurs at Digantara to

Working on a CubeSat at Digantara

tackle the incredibly hard problem of space debris. The ideas are really innovative to say the least. Instead of mostly ground based tracking of space debris, Digantara hopes to deploy debris sensing satellites in orbit, "our 'Inorbit Space Debris Monitor' is a Hardware-Software combined technology for Debris detection and modelling. We deploy a constellation of satellites equipped with debris detection modules to track and map the space objects. Given that space-based sensors do not suffer atmospheric losses, we will be able to detect much smaller objects. The obtained data is processed on ground to model the debris environment and deliver SSA services for customers involved in satellites building & operations, launch, insurance, regulation, debris mitigation and military." There are all kinds of junk in orbit around the Earth, and Digantara hopes to focus on the smaller pieces that are not tracked by ground based technologies, "we are focused on mid-sized debris which pose high threat to active satellites but are too small to be tracked by ground based systems. We will be able to track 18 times more objects including large objects that the ground-based systems track."

The technology will also be used to monitor and assess the threat of collisions, so that we do not get a Kessler Cascade like the one that takes place in the movie Gravity, "the increasing threat of collisions in space made Space Situational Awareness (SSA) a focus area in many countries. Ground based technologies like RADARs, Optical and Radio telescopes form today's debris detection capabilities. They suffer greatly from atmospheric distortion thereby creating a need for space based surveillance platforms owing to the increased quality of data it can deliver. Space companies have also been developing solutions to remove the large debris in space."

BELBIRD TECHNOLOGIES OFFERS SURVEILLANCE... WITHOUT CAMERAS

BelBird technologies offers perimeter protection and intruder detection for office spaces and homes, without using security cameras! The whole system work on device IDs and sensors, "the surveillance of a particular space through presence sensors and mobile handset attributes like MAC IDs in that space are collected and mapped together. The BelBird combo sensor is compact, miniature, and cost-effective in that it could be deployed in every nook and corner. BelBird has come out with another product case where it will be incorporated in lighting units and an integrated unit will use lighting also